



## Koneru Lakshmaiah Education Foundation

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### OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

#### Project Problem Statement Submission Form

##### Section / Team Details

Section No. : 10

Team No. : 11

Project Title: Alarm clock

##### Team Members

| Roll Number | Student Name       | Signature              |
|-------------|--------------------|------------------------|
| 2520090183  | M.Priyavallika     | <i>M. Priyavallika</i> |
| 2520080033  | D.Shamika Hasini   | <i>Shamika Hasini</i>  |
| 2520090157  | SamyukthaKandukuri | <i>Samyuktha</i>       |

##### 1. Abstract

The Alarm Clock project is a Linux-based application developed to schedule and trigger alarms at specified times. The main objective is to provide a simple alarm system while demonstrating important Operating Systems and system programming concepts. The application allows users to set, view, and cancel alarms and notifies the user when the scheduled time is reached.

The project focuses on Linux/POSIX mechanisms such as processes, signals, timers, and system calls. Functions such as `fork()`, `clock_gettime()`, `nanosleep()`, and `sigaction()` can be used to monitor time and handle alarm events. These mechanisms demonstrate how Linux manages processes, timing, and asynchronous signals.

The application will run in a Linux environment and use a command-line interface. The expected outcome is a functional and reliable Alarm Clock that performs its basic purpose while clearly demonstrating Operating Systems concepts and Linux system programming techniques.

## **2. Problem Statement**

The main problem addressed by this project is the need for a simple and reliable system to schedule and trigger alarms at a specified time. The project aims to develop a Linux-based Alarm Clock that can accurately monitor the system time and notify the user when the set alarm time is reached.

The project is relevant to Operating Systems and Systems Programming because it uses Linux/POSIX features such as processes, signals, timers, and system calls. It demonstrates how a program interacts with the operating system to manage time-based events and handle asynchronous notifications. The project therefore provides practical understanding of process management, signal handling, and system-level programming in a Linux environment.

## **3. Objectives**

1. To develop a Linux-based Alarm Clock that allows users to set and manage alarms easily.
2. To accurately monitor the system time and trigger an alarm at the scheduled time.
3. To demonstrate Linux/POSIX concepts such as processes, signals, timers, and system calls.
4. To provide a reliable and simple command-line interface for setting, viewing, and cancelling alarms.

## **4. Proposed Methodology**

The project will be implemented in a Linux environment using C and POSIX system calls. The user will enter the required alarm time through a simple command-line interface. The program will obtain the current system time and calculate the required waiting period.

A separate process or timer mechanism will be used to monitor the alarm without affecting the main program. Linux functions such as `fork()`, `clock_gettime()`, `nanosleep()`, and `sigaction()` will be used for process creation, time monitoring, waiting, and signal handling. When the scheduled time is reached, a signal will be generated and the alarm notification will be displayed.

The basic flow will be:

**set alarm → calculate time → wait/monitor → trigger signal → notify user → stop or reset alarm.**

The program will also provide options to view or cancel the active alarm.

## 5. Operating Systems Concepts / Linux APIs Used

| OS Concept / Linux API / System Call | Purpose in the Project                                              |
|--------------------------------------|---------------------------------------------------------------------|
| fork()                               | Creates a separate process to monitor the alarm.                    |
| Clock_gettime()                      | Gets the current system time to compare with the alarm time.        |
| nanosleep()                          | Makes the monitoring process wait efficiently until the alarm time. |
| Sigaction()/Signals                  | Handles the alarm event and triggers the notification.              |
| Process Management                   | Manages the alarm-monitoring process and its execution.             |
| POSIX Timers                         | Provides accurate time-based scheduling for triggering the alarm.   |

## 6. Individual Contribution

| Roll Number | Student Name       | Individual Responsibility                                                                |
|-------------|--------------------|------------------------------------------------------------------------------------------|
| 2520090183  | M.Priyavallika     | Design the alarm clock interface and implement alarm setting, viewing, and cancellation. |
| 2520080033  | D.Shamika Hasini   | Implement time monitoring, POSIX timers, and alarm scheduling using Linux system calls.  |
| 2520090157  | SamyukthaKandukuri | Implement signal handling and alarm notification, and perform testing and debugging.     |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                                                                              |
|----------------------------|--------------------------------------------------------------------------------------|
| Linux / Ubuntu             | Provides the operating system environment for developing and running the alarm clock |
| C / C++                    | Used to implement the main Alarm Clock program and system-level functionality        |
| GCC Compiler               | Compiles the C/C++ source code into an executable program                            |
| POSIX Libraries            | Provides APIs for processes, signals, timer, and time management                     |
| Visual Studio Code         | Used for writing and editing the source code                                         |

### \* Expected Outcome

The project is expected to provide a simple and reliable Linux-based Alarm Clock that allows users to set and manage alarms. The system will monitor the current time and trigger an alarm notification when the scheduled time is reached. It will also demonstrate important Linux/POSIX Operating Systems concepts through practical implementation.

1. Set, view, and cancel alarms.
2. Accurately monitor system time.
3. Trigger an alarm at the scheduled time.
4. Use Linux processes, signals, timers, and system calls.
5. Provide a simple command-line interface.
6. Demonstrate practical Operating Systems and system programming concepts.

### 9. GitHub Repository

GitHub Repository Link : shamikahasini-27/OSSP-project

Repository Created : Yes

### Faculty Name and Signature

Faculty Name: Raghupathi.M

Signature: \_\_\_\_\_

Remarks: \_\_\_\_\_

### Faculty Use

Project Approved: \_\_\_\_\_

☒ Yes ☐ No

Date: \_\_\_\_\_

Faculty Signature: \_\_\_\_\_